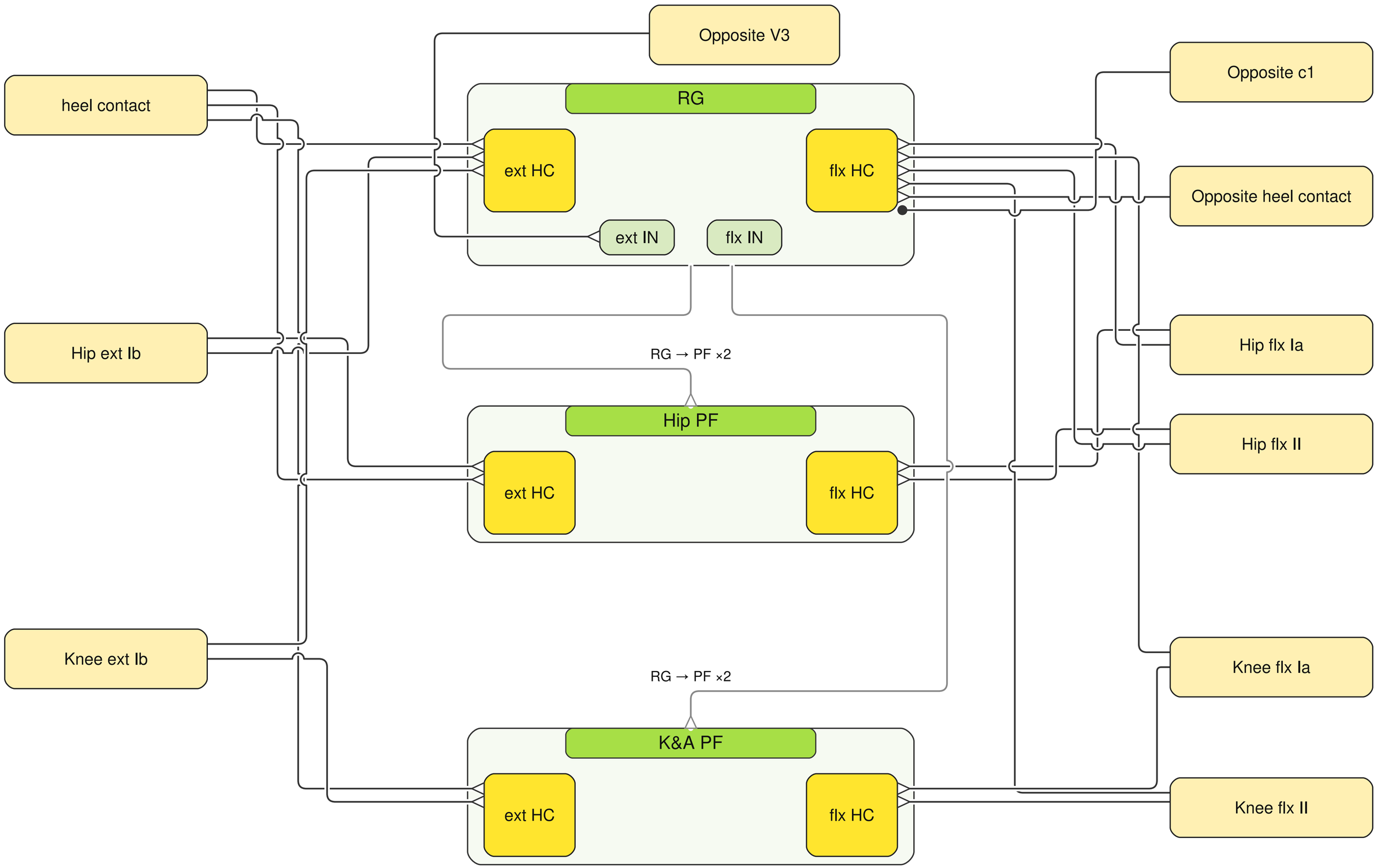
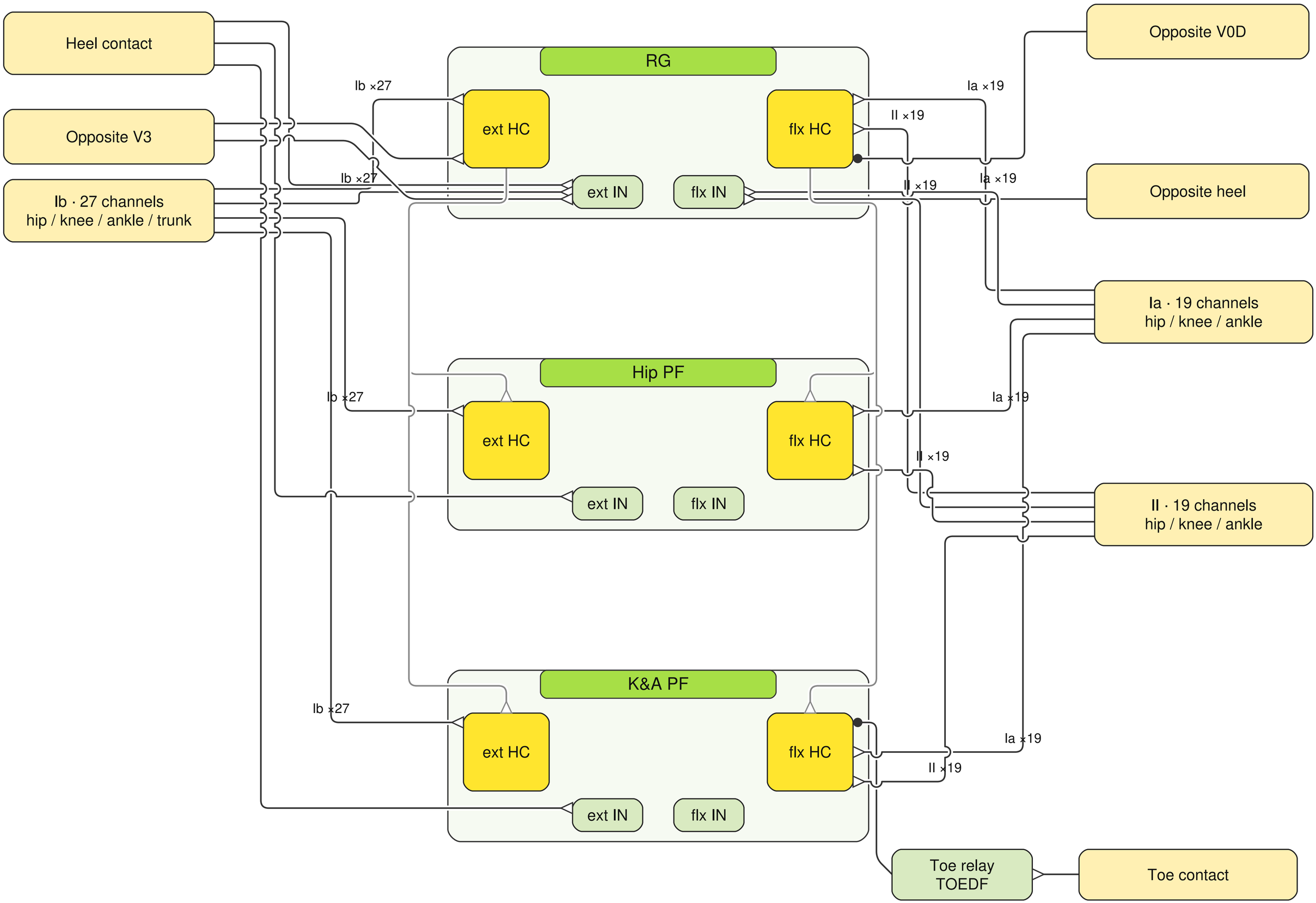


Excitation: triangle base on destination · Inhibition: filled circle · Crossings: semicircular bridges

Internal RG/PF wiring omitted. Opposite-side sources are placed beside their RG destinations.

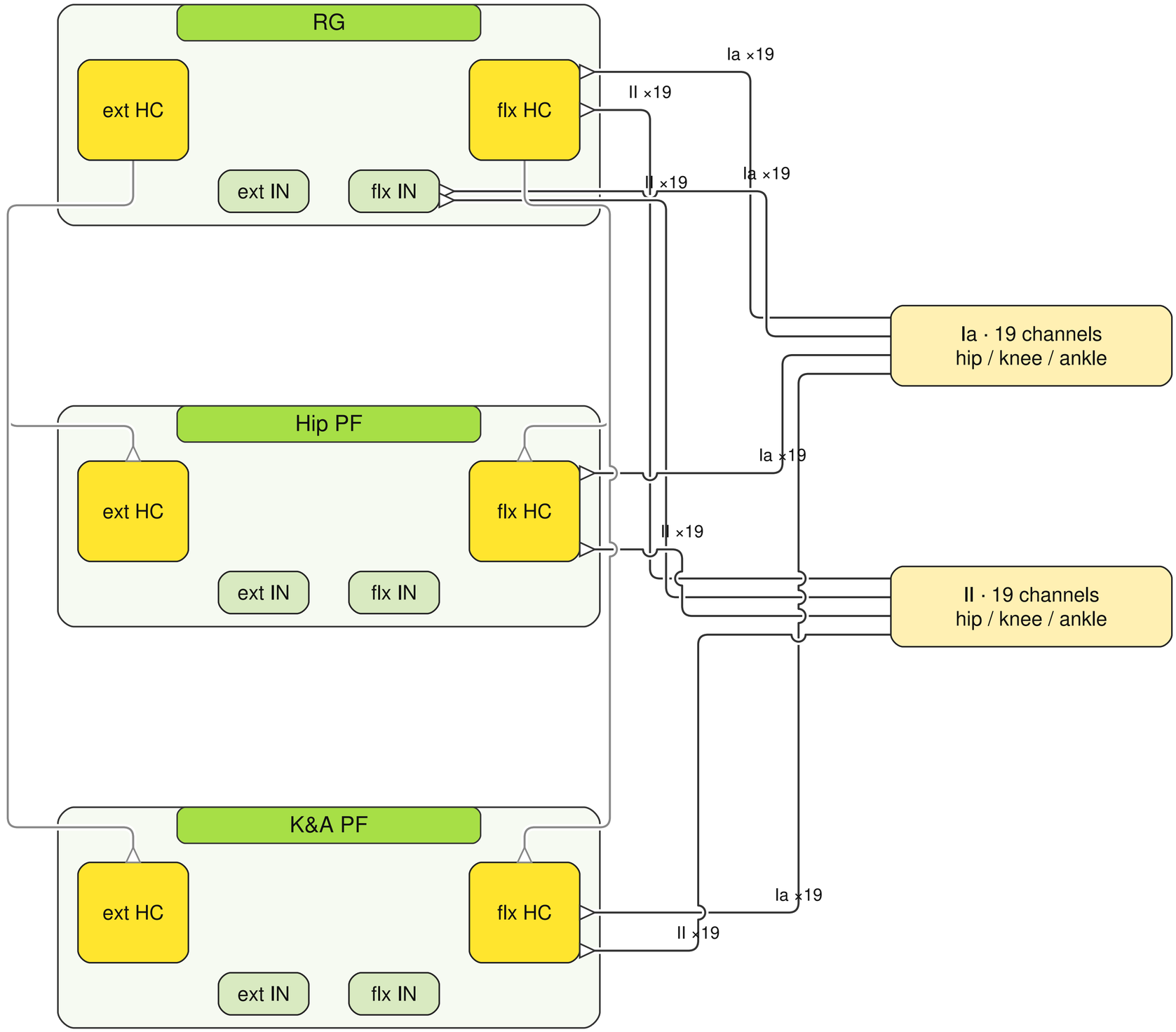
w2lvar · RG, Hip PF, and K&A PF



×N = separate muscle-channel synapses; their individual destination parameters are retained in the key.

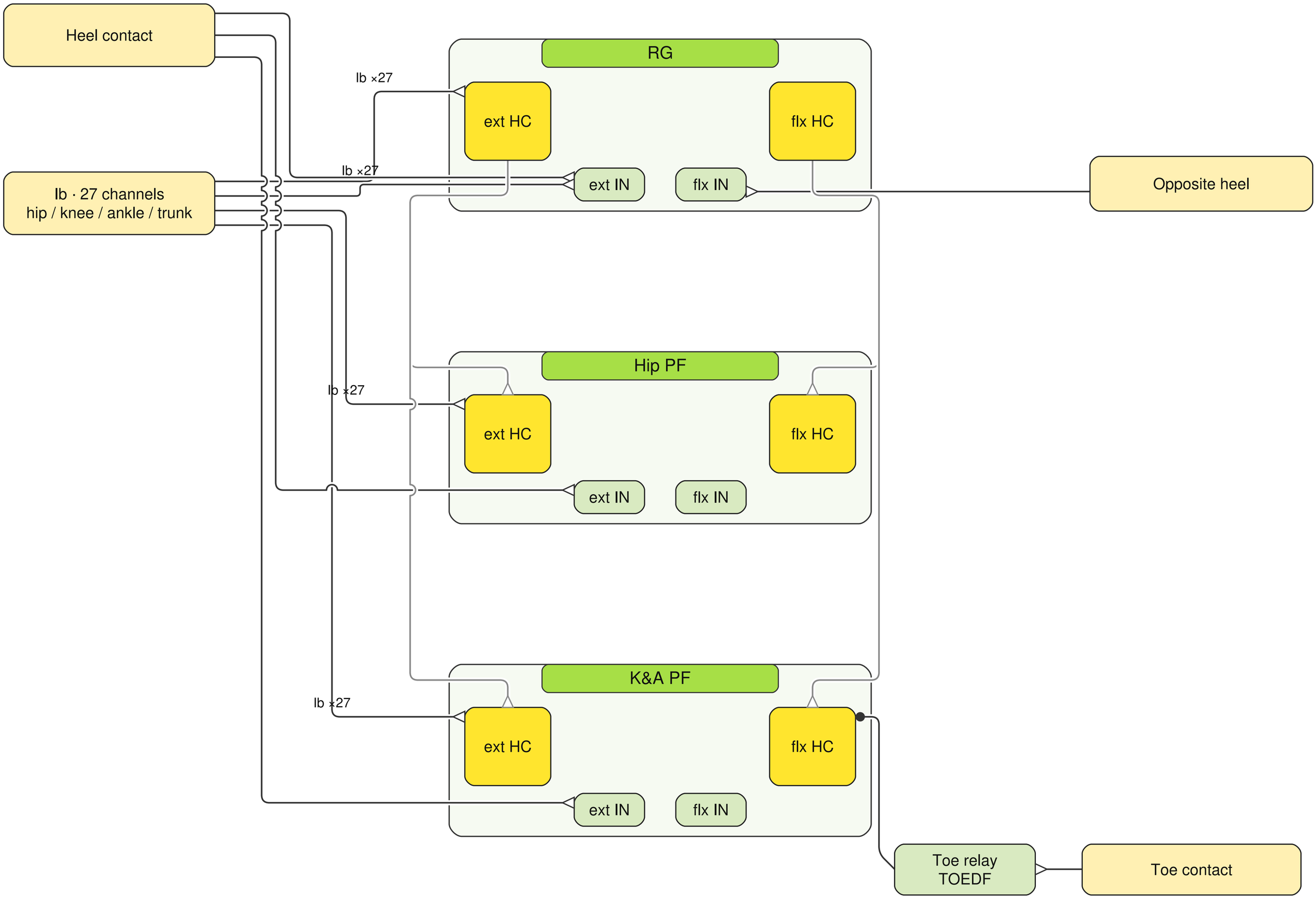
K&A is the shared knee–ankle PF pair. This page uses one set of destination objects, without duplicate HCs.

w2lvar · Ia and II pathways



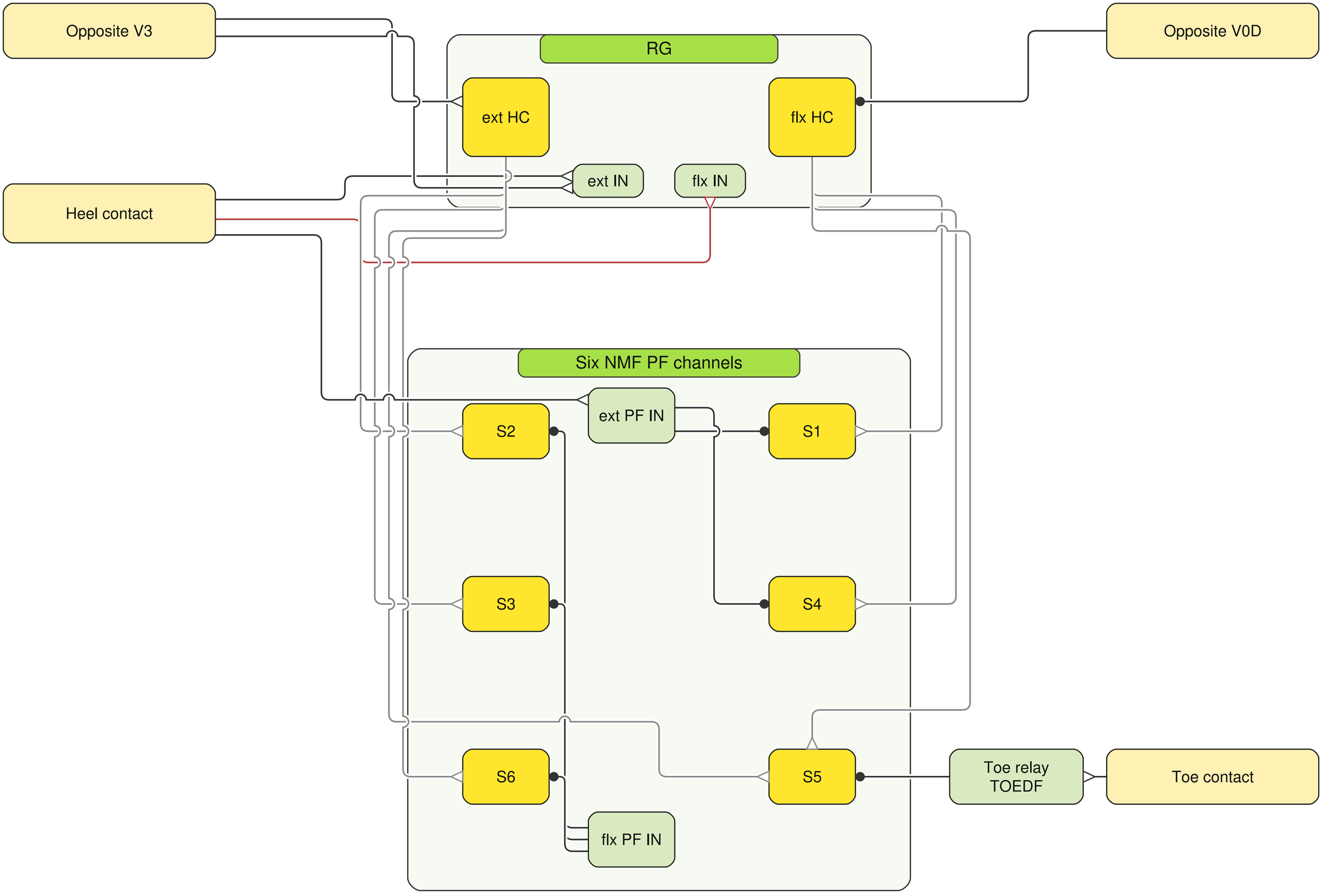
Same RG / Hip PF / K&A PF arrangement as the overview. Gray lines retain the RG→PF context.

w2lvar · lb, heel, and toe pathways



Same RG / Hip PF / K&A PF arrangement as the overview. Gray lines retain the RG→PF context.

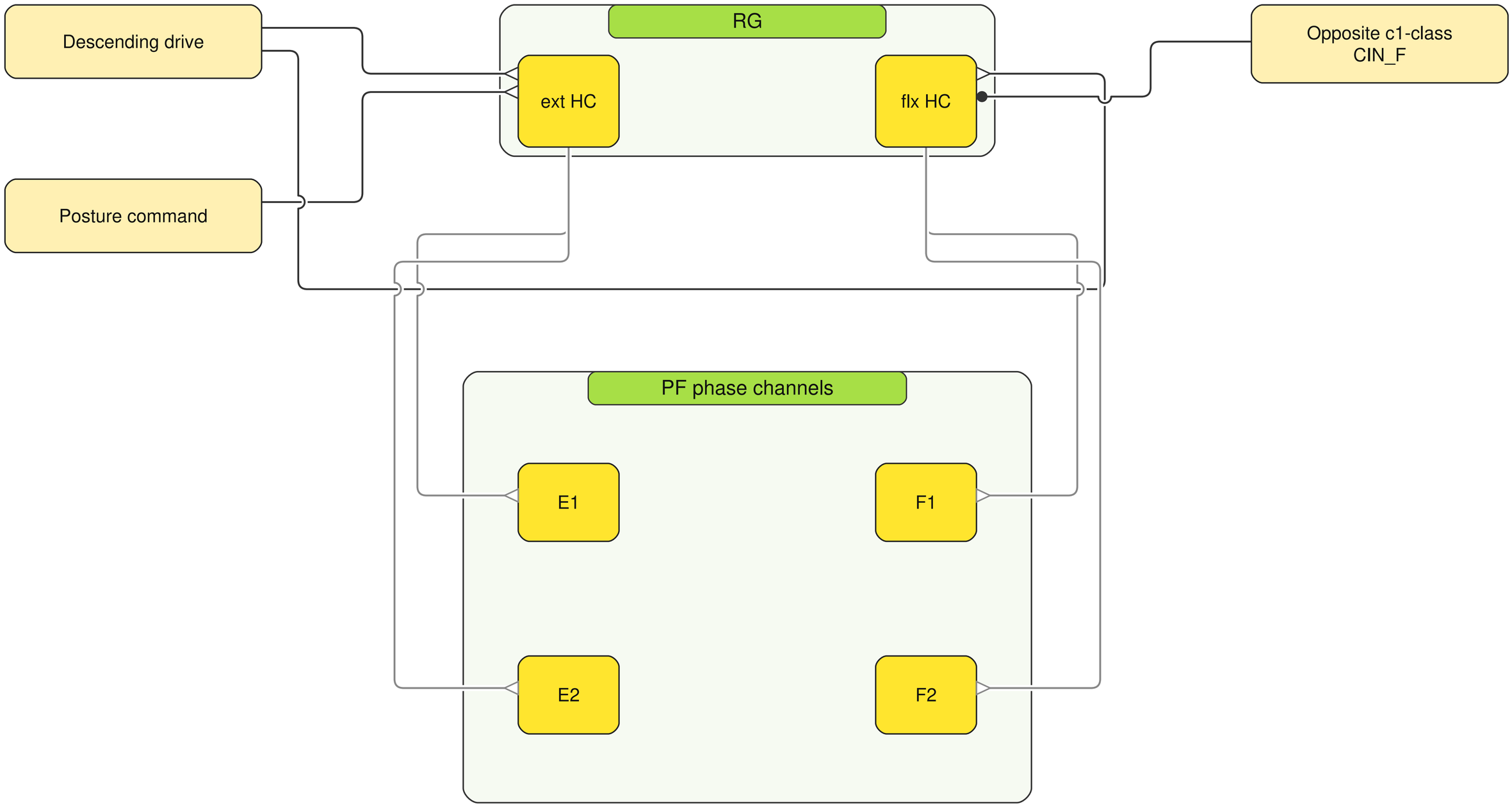
syn6 · RG and six NMF pattern-formation channels



Red dashed: the saved heel→flexor-IN branch disputed in Comment 30; the source circuit has not been changed.

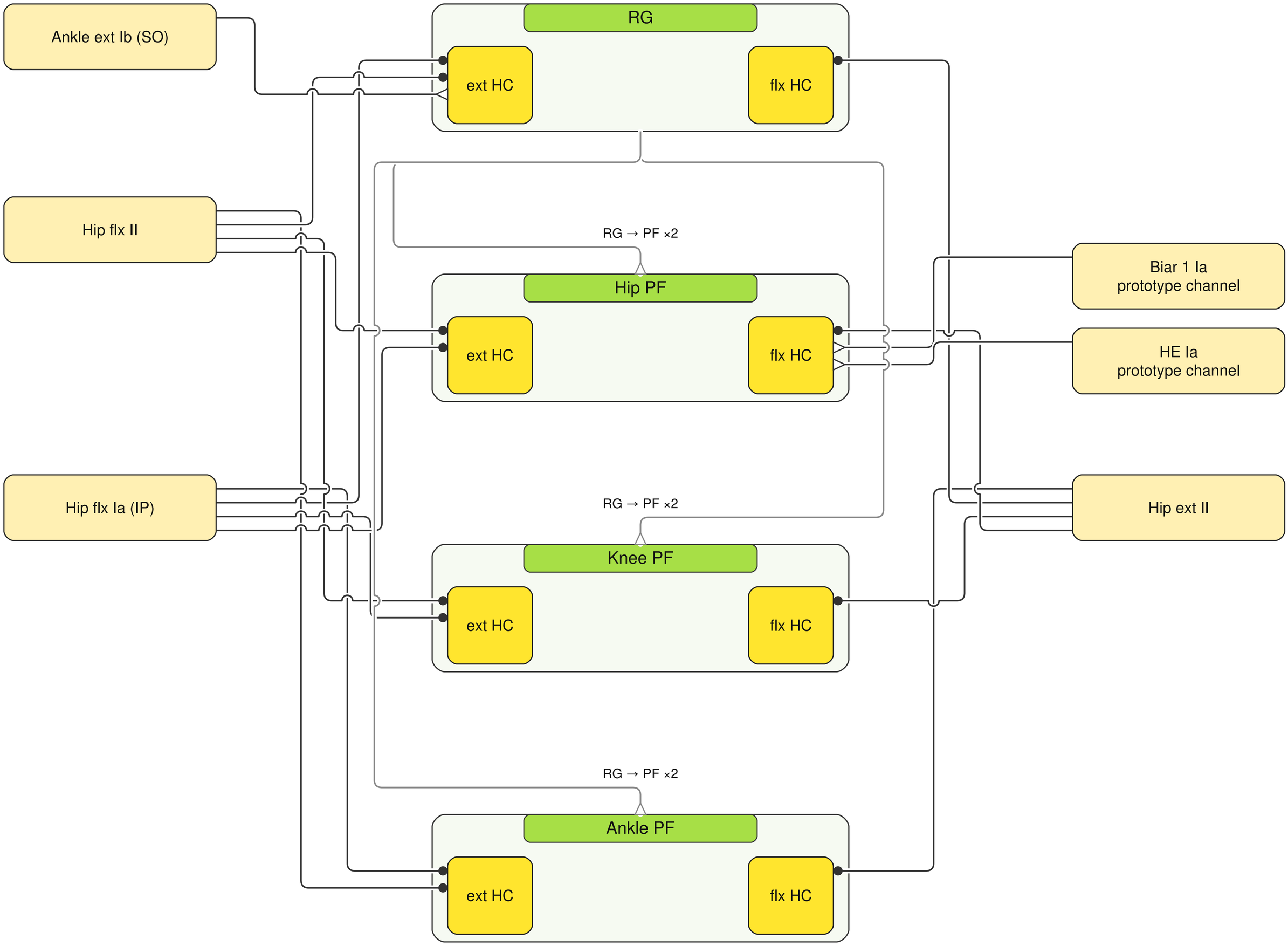
S1–S6 are PF channels, not hip/knee/ankle pairs. Saved contact-input gains are zero in this standing condition.

s3k · RG and four PF phase channels



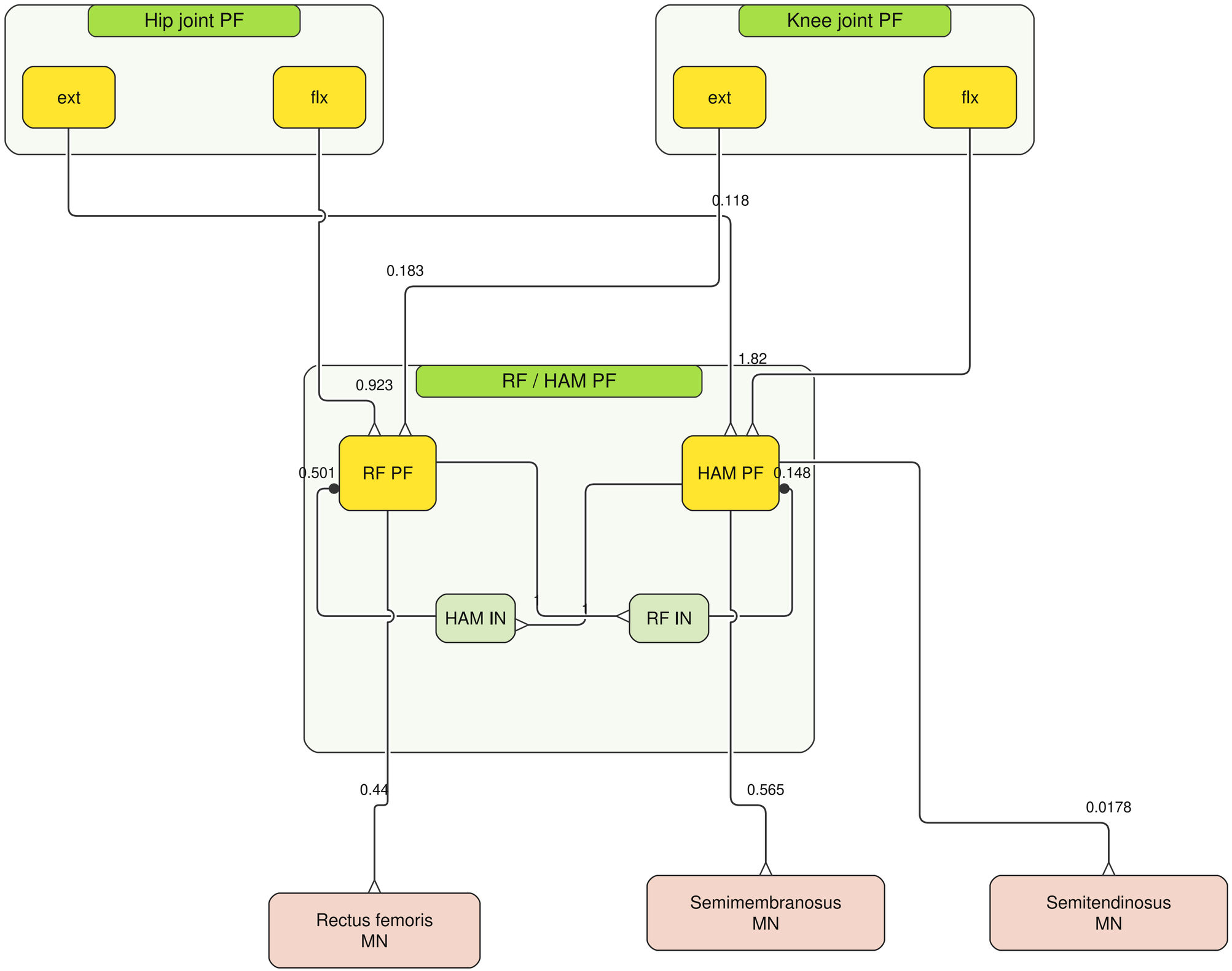
Saved standing condition: actual descending, contralateral, and RG→PF connections. Internal motifs omitted.

Rat134 · curated receptor-geometry condition

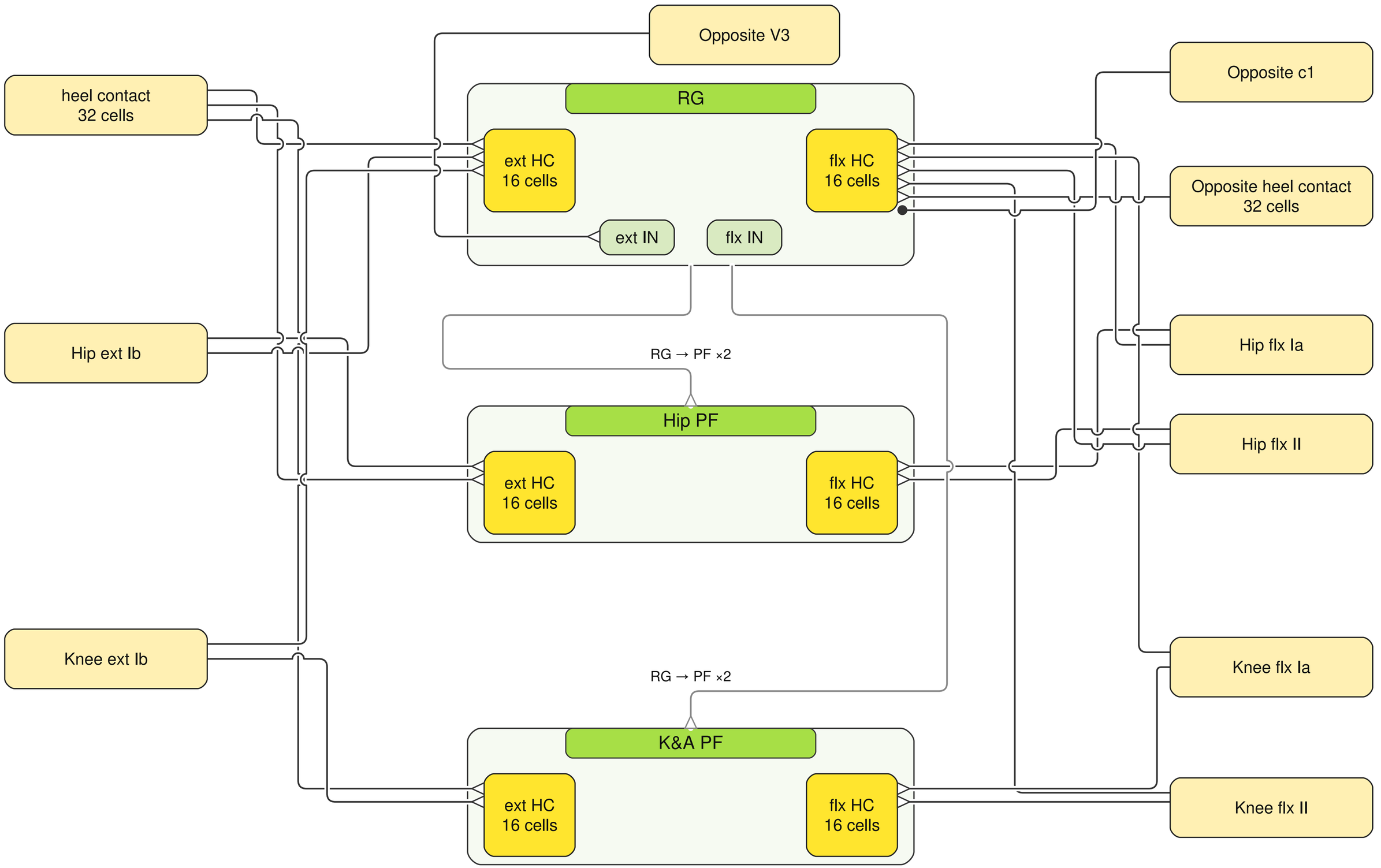


Separate knee and ankle PF modules are retained here. These are source connections, not assumed human-model pathways.

RF / HAM · four-cell PF subnetwork (saved left-side projection)

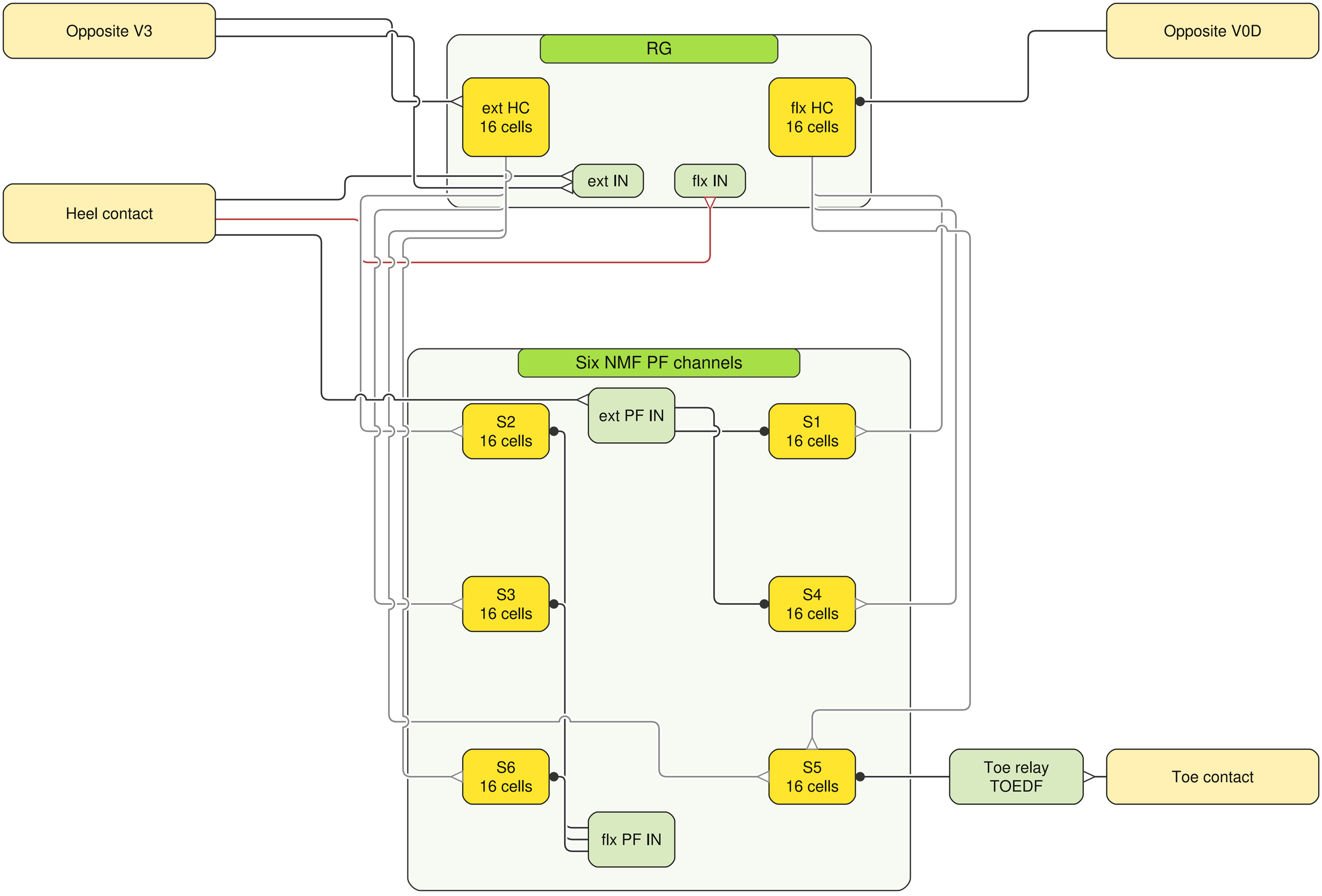


Joint PF → RF/HAM pair → motor neurons. Expanded four-cell connections shown; other afferents are outside this saved projection.



Population readouts drive the chemical pathways; these are not soma-to-soma spike-event arrows. Internal motifs omitted.

Gait2392 TimingZeroSFA · central spiking populations and PF channels



RG and S1–S6 are spiking populations; INs remain graded. Chemical pathways use spike-derived proxy voltage. Red branch remains disputed.

Source / destination / synapse: what the labels mean

II_tib_ant names one type-II afferent neuron, not one globally fixed synapse.

A connection is defined by its source, destination, conductance, reversal, and presynaptic window.

The same afferent can therefore have different parameters at different destinations.

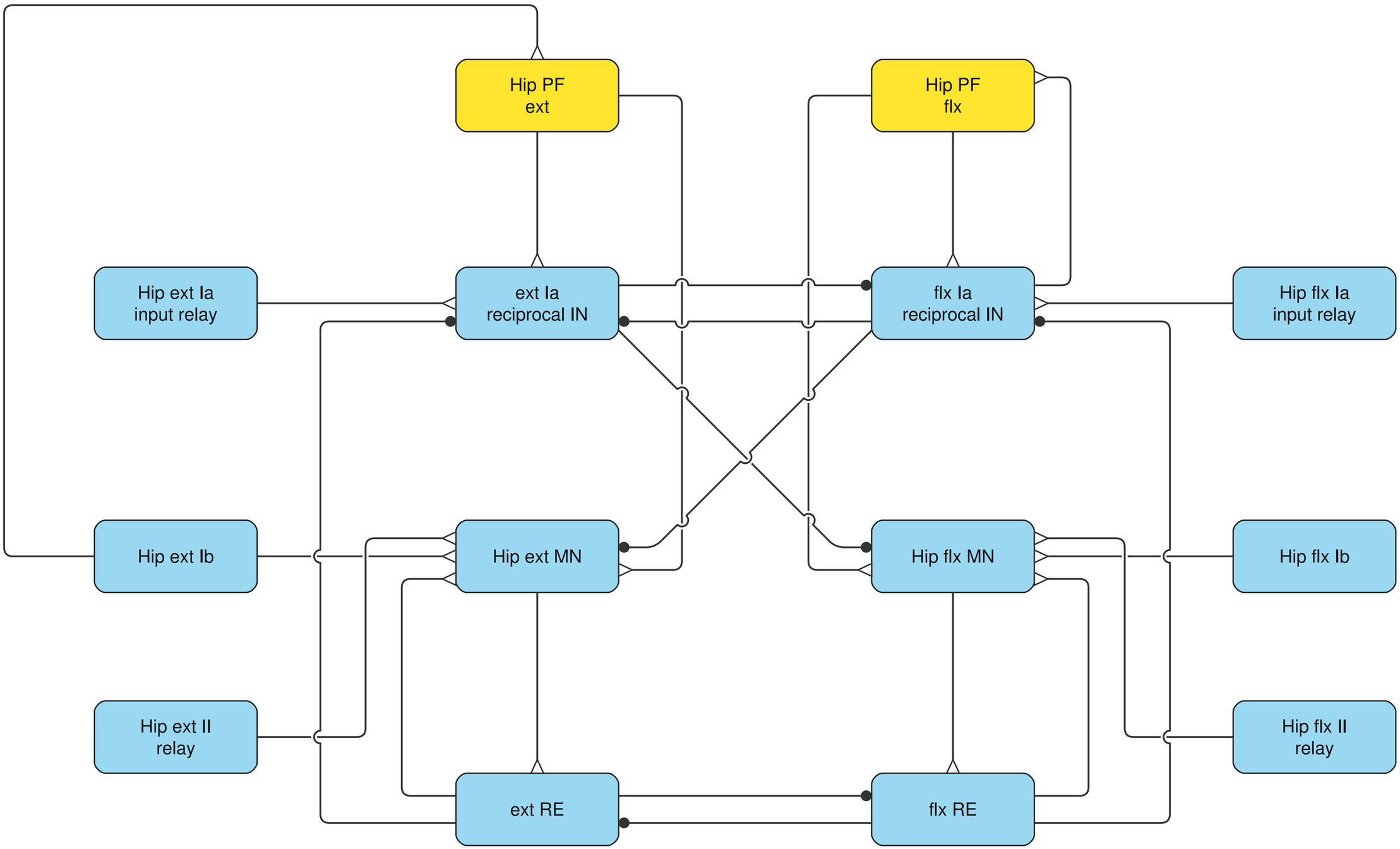
The checked left/right homologues below match; diagram labels omit the side suffix.

Source → destination	g (uS)	Erev (mV)	lo / hi (mV)	L = R
II_tib_ant → RG_F	0.0278947	8.0	0 / 5	yes
II_tib_ant → InF	0.0278947	8.0	0 / 5	yes
II_tib_ant → PF_HIP-F	0.0278947	8.0	0 / 5	yes
II_tib_ant → PF_KNEE-F	0.0278947	8.0	0 / 5	yes
II_tib_ant → IIX_tib_ant	1	8.0	0 / 5	yes
II_tib_ant → IIIN_tib_ant	1	8.0	0 / 5	yes
IIIN_tib_ant → MN_med_gas	0.133333	-5.0	0 / 5	yes
IIIN_tib_ant → MN_lat_gas	0.133333	-5.0	0 / 5	yes
IIIN_tib_ant → MN_soleus	0.133333	-5.0	0 / 5	yes
IIIN_tib_ant → MN_tib_post	0.133333	-5.0	0 / 5	yes
IIIN_tib_ant → MN_flex_dig	0.133333	-5.0	0 / 5	yes
IIIN_tib_ant → MN_flex_hal	0.133333	-5.0	0 / 5	yes
IIX_tib_ant → MN_tib_ant	0.4	8.0	0 / 5	yes
IIIN_tib_ant → MN_per_brev	0.133333	-5.0	0 / 5	yes
IIIN_tib_ant → MN_per_long	0.133333	-5.0	0 / 5	yes
IIIN_tib_ant → MN_per_tert	0.133333	-5.0	0 / 5	yes

A ×N line bundles N separately parameterized connections; it is not a single summed synapse.

Ankle channels are included in the muscle bundles where present. Complete members and edge IDs accompany the drawings.

1 · Paired MN circuit: the saved W2L108 hip subnetwork



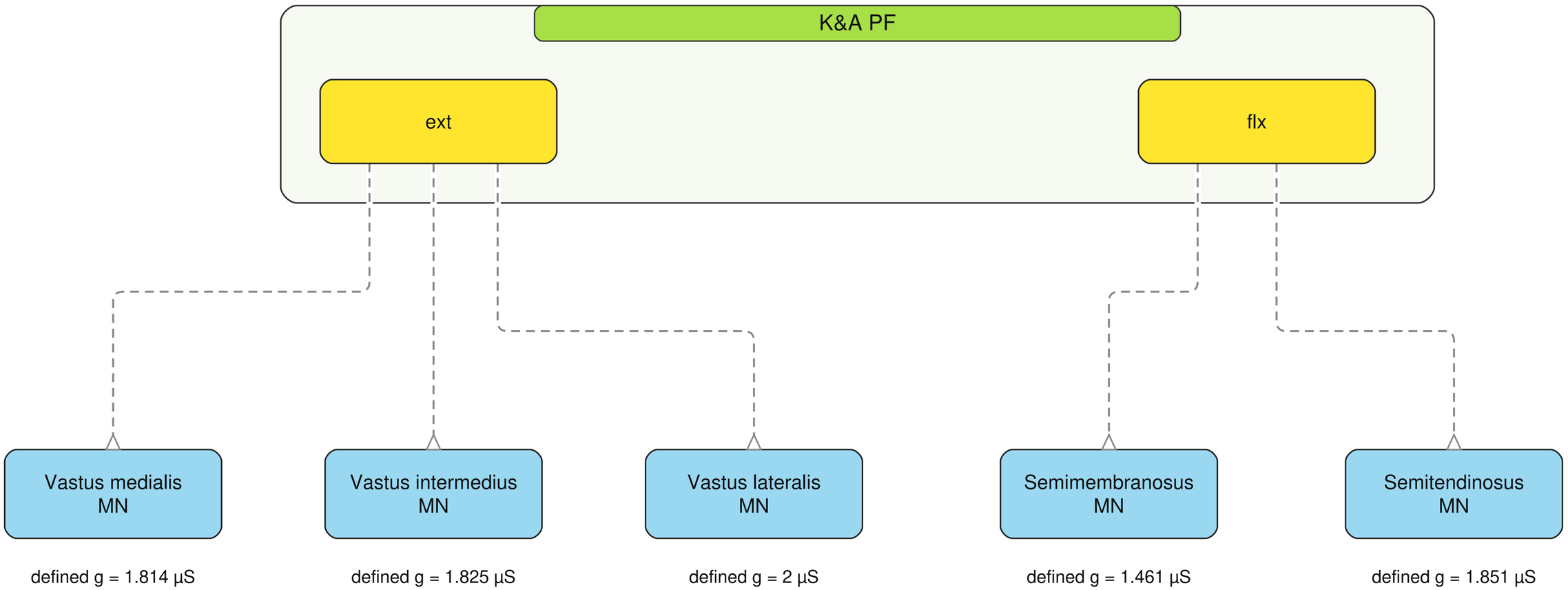
One extensor MN and one flexor MN; all saved neural connections among the displayed cells are shown.

Native naming: the peripheral “Ia IN” relays feed the reciprocal-inhibition cells named “Ia”.

Source detail: RE → own MN is excitatory here. RE → Ia and RE → opposite RE are inhibitory.

Open triangle: excitation (base on target). Filled circle: inhibition. Semicircular bridges: no junction.

2 · Several muscle MNs share each side of a PF pair



These are five separate motor pools, not one pooled “extensor” or “flexor” MN.

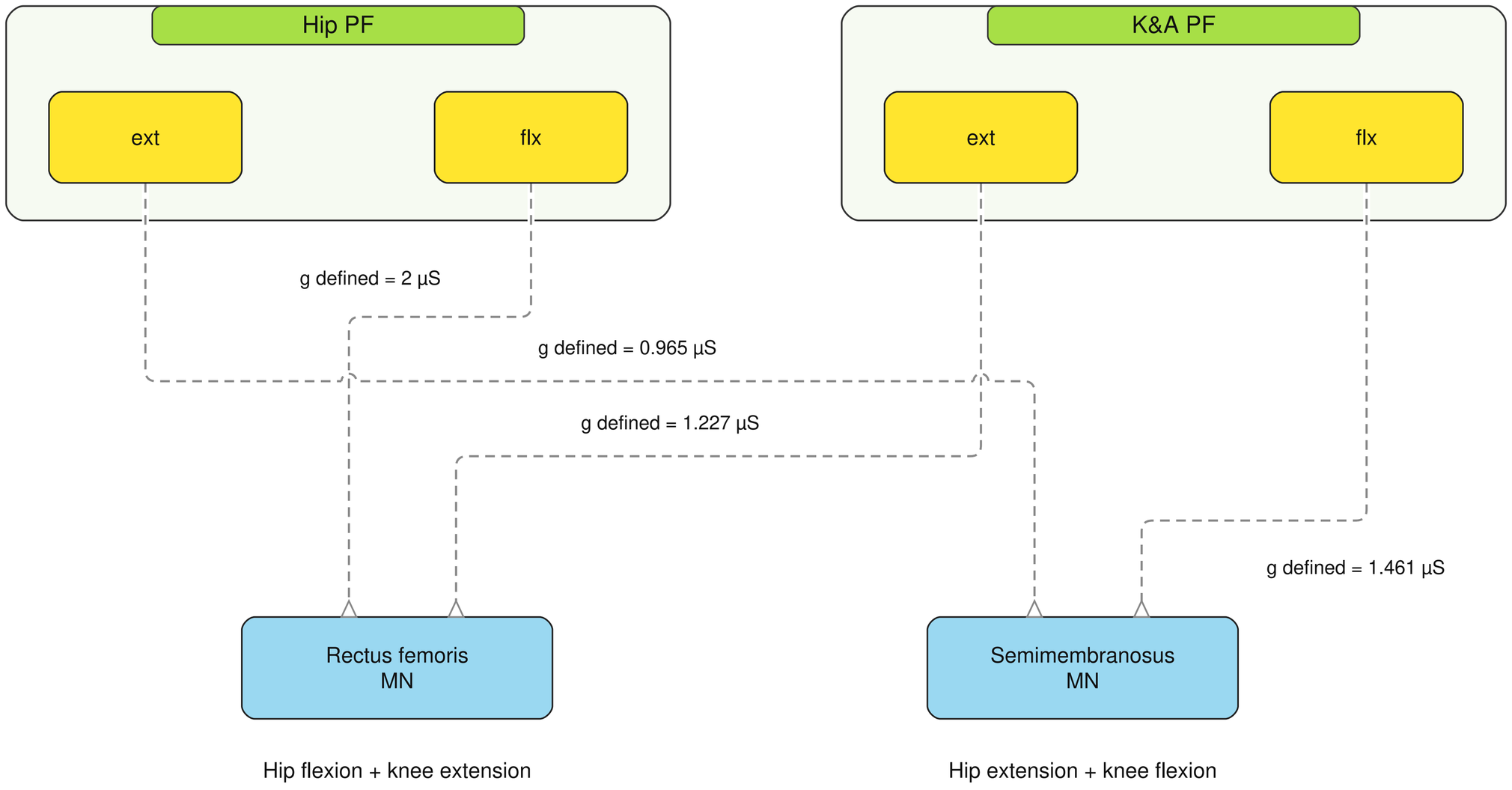
Each has its own afferents and interneurons; sharing a PF source does not merge those circuits.

Selected w2Ivar fan-out. Additional PF inputs to the biarticular hamstrings are shown in example 3.

Gray dashed = source-defined connection with effective g = 0 in this saved standing condition.

K&A is the model’s combined knee–ankle PF pair; only the selected knee-muscle branches are expanded here.

3 · A biarticular muscle receives separate inputs from two PF layers



Each muscle has ONE MN pool. Both PF synapses terminate on that same pool.

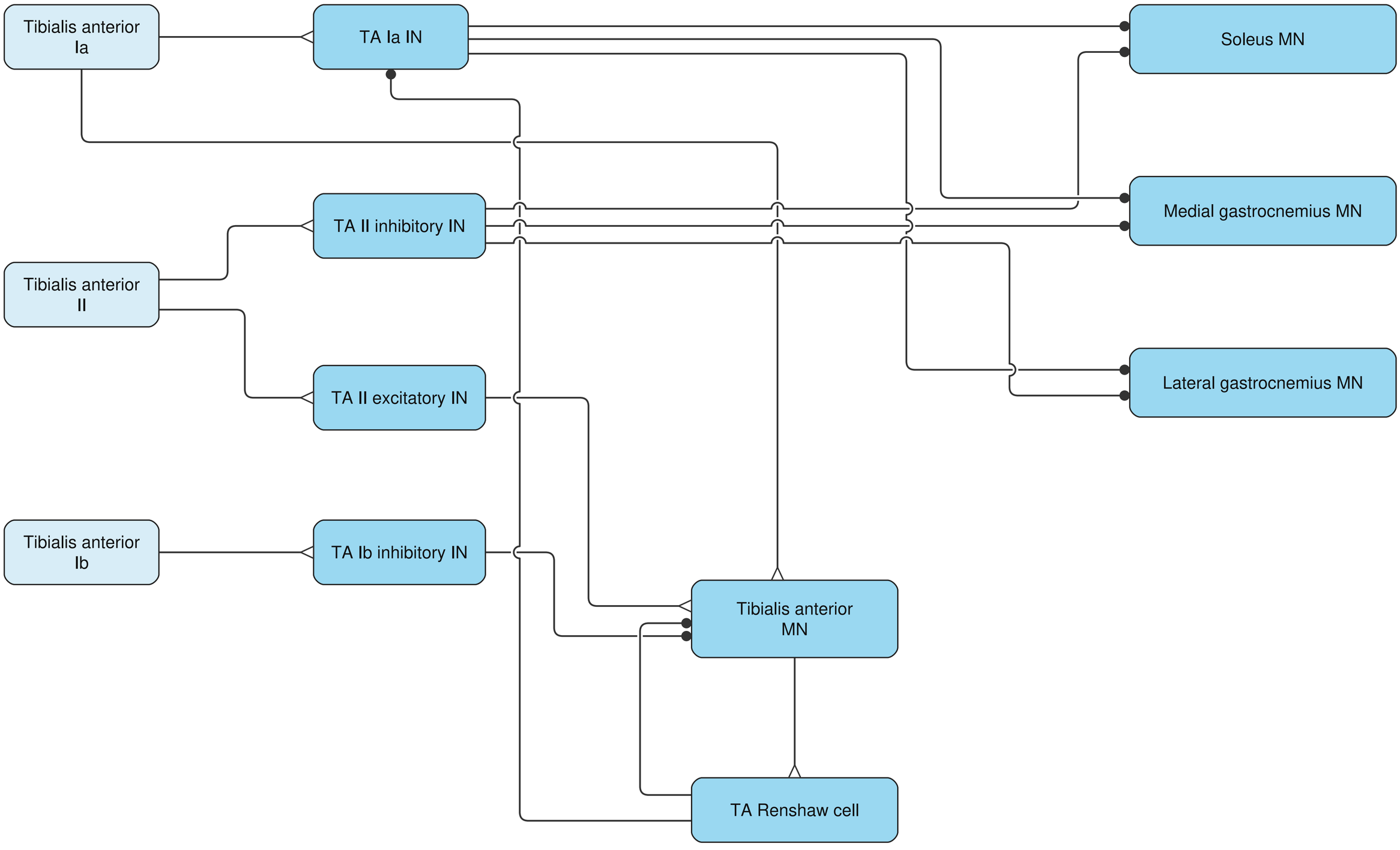
Both inputs are excitatory; separate conductances contribute to the MN membrane equation.

A muscle's action at two joints does not require a second MN or a second copy of its local reflex circuit.

Gray dashed = defined source routing; every displayed PF→MN edge has effective g = 0 in the saved w2Ivar condition.

Weights belong to individual source→target synapses. Sharing an MN does not make the two PF connections one synapse.

4 · One muscle’s local motif can inhibit several antagonist MNs



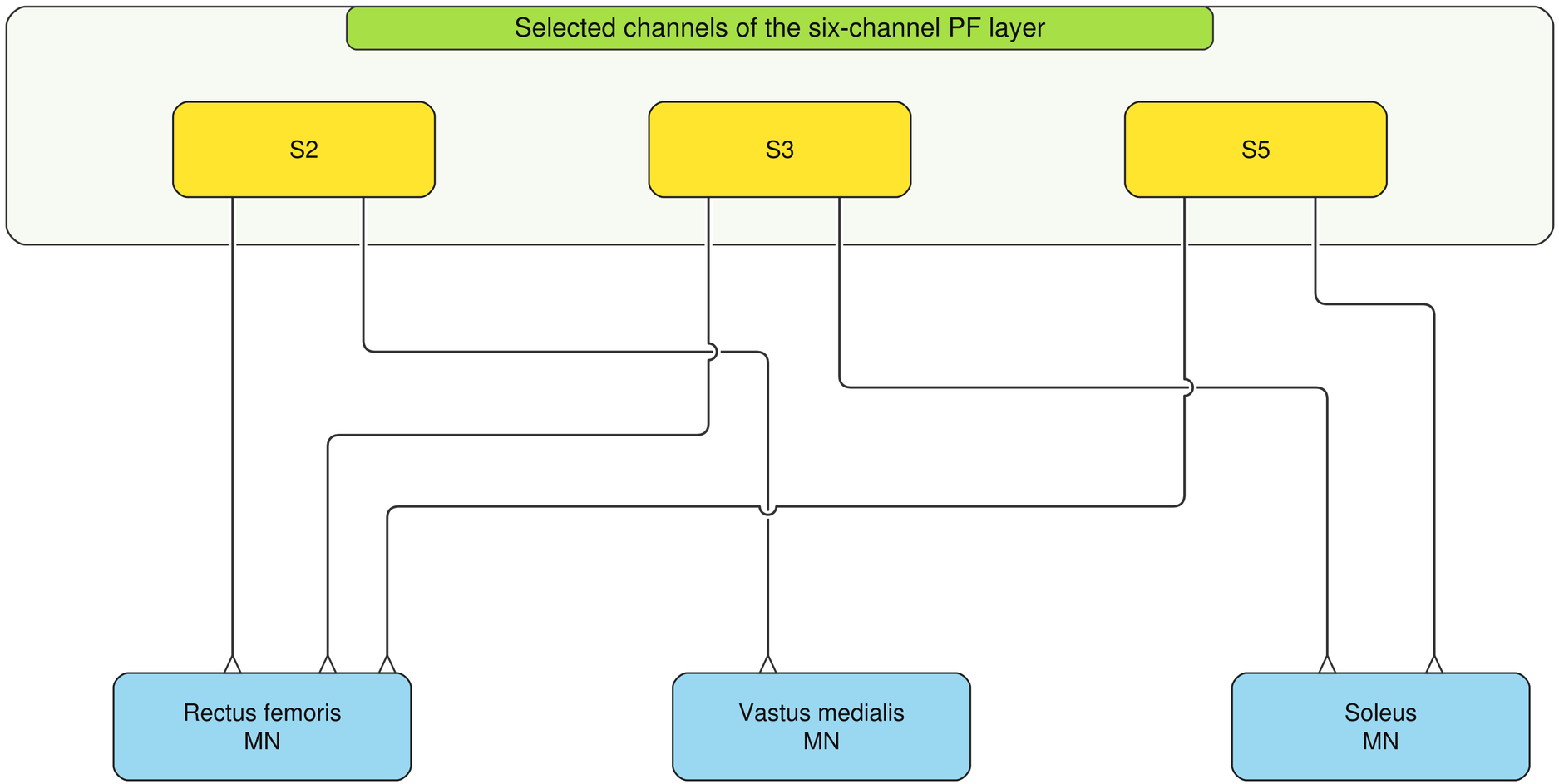
A separate inhibitory synapse reaches each antagonist MN; the three targets are not one pooled muscle.

The same TA Ia neuron also excites its own MN. II uses separate excitatory and inhibitory interneurons.

PF gating, central feedback, other antagonist muscles, and the wider Renshaw network are outside this focused view.

TA = tibialis anterior. All lines on this page are nonzero saved w2lvar connections. No muscle or MN is duplicated.

5 · Active syn6 PF channels converge and fan out across muscles



All three PF inputs to rectus femoris are shown. The other two MNs also receive inputs from channels outside this view.

Rectus femoris integrates three separate PF synapses; S2 also branches to vastus medialis.

S3 and S5 also branch to soleus. These channels encode muscle synergies, not named hip/knee half-centers.

S2 → rectus femoris: g = 0.007152 µS; S2 → vastus medialis: g = 0.01747 µS.

Each outgoing synapse has its own weight. Other channel inputs and local reflex circuits are omitted here.

All displayed connections are excitatory and nonzero. Exact conductances and thresholds accompany the editable drawing.